1 Step 7: Duration-Aware, Lagged ROI Analysis with Variable Trial Windows

1.1 Objective

The objective of Step 7 is to replace the fixed post-onset summary window used in Steps 5 and 6 with a **question-specific variable analysis window** that depends on the actual duration of each trial.

The goal of this step is to test whether the abstract-versus-concrete effect in the left frontal ROI remains when the fNIRS summary is computed over a trial window that:

- starts from the actual question onset;
- ends at the actual participant answer;
- includes a haemodynamic lag;
- varies from trial to trial instead of being fixed at the same length for every question.

This step should remain simple. It should produce:

- one primary covariate-adjusted inferential model;
- exact p-values;
- confidence intervals;
- effect estimates;
- a short final conclusion.

1.2 Why Step 7 Is Needed

Step 6 showed that the fixed-window front ROI analysis was close to, but still above, the conventional significance threshold. At the same time, the descriptive visualizations suggested that:

- condition differences evolve over time rather than appearing in one perfectly fixed segment;
- the waveform shape varies across channels;
- the questions differ in length, structure, and difficulty.

Therefore, a variable-window analysis is a natural next step because it allows the fNIRS summary to follow the real duration of the question-processing period for each trial.

1.3 Core Analytical Decision

To keep Step 7 interpretable, the following elements should remain fixed from Step 6:

1. the same preprocessed continuous fNIRS files from Step 3;
2. the same included participant-session cohort carried forward from Step 6;
3. the same long-channel-only rule;
4. the same frontal ROI as the primary ROI;
5. the same short-channel exclusion from inference;
6. the same question-level covariates;
7. the same mixed-effects modeling framework with participant and question random intercepts.

The main change in Step 7 is:

> the trial outcome is no longer based on a fixed 7–11 s window, but on a lagged question-specific interval that varies with the actual trial duration.

1.4 Primary Scientific Question

The primary Step 7 question is:

> In the left frontal ROI, does the abstract-versus-concrete effect remain after adjusting for question length and item difficulty when the fNIRS response is summarized over a duration-aware, lagged, trial-specific interval rather than a fixed window?

1.5 Why Continuous Preprocessed Data Must Be Used

Step 5 and Step 6 were based on onset-locked epochs extending from:

```
-3.5 s to 13.2 s around question onset
```

That epoch structure was appropriate for fixed-window analyses. However, Step 7 should not rely on those fixed-length epochs as the main analysis input, because a variable trial window may extend beyond the end of that epoch for slower responses.

Therefore, Step 7 should use:

> the continuous preprocessed raw data from Step 3 together with the harmonized trigger table.

1.6 Primary ROI Definition

The primary ROI remains the same left frontal ROI used in Steps 5 and 6.

1.6.1 Front ROI HbO channels

```
S2_D6 hbo
S2_D4 hbo
S1_D6 hbo
S1_D3 hbo
S5_D6 hbo
S5_D4 hbo
S5_D3 hbo
```

1.6.2 Back ROI HbO channels

The back ROI is retained as a secondary ROI:

```
S4_D2 hbo
S3_D1 hbo
S6_D2 hbo
S6_D1 hbo
S6_D5 hbo
S7_D2 hbo
S7_D5 hbo
S8_D1 hbo
S8_D5 hbo
```

1.6.3 Transition channels excluded from the primary confirmatory summaries

```
S5_D7 hbo
S4_D4 hbo
S4_D7 hbo
S3_D3 hbo
S3_D7 hbo
S6_D7 hbo
```

1.6.4 Short-separation pairs excluded from inference

```
S1_D8
S2_D9
S3_D10
S4_D11
S5_D12
S6_D13
S7_D14
S8_D15
```

1.7 Primary Trial Window Definition

For participant i and trial k, let:

- t_{ik}^{on} denote the harmonized question onset time;
- t_{ik}^{ans} denote the harmonized answer time;
- t_{ik}^{next} denote the next question onset time, if it exists.

The raw behavioral duration of the trial is:

$$L_{ik} = t_{ik}^{\text{ans}} - t_{ik}^{\text{on}}. \tag{1}$$

The primary Step 7 analysis window should be a **lagged duration-aware interval**:

$$W_{ik}^{(\delta)} = \left[t_{ik}^{\text{on}} + \delta,\ \min\!\left(t_{ik}^{\text{ans}} + \delta,\ t_{ik}^{\text{next}} - 0.5\right)\right], \tag{2}$$

where the primary lag is:

$$\delta = 4 \text{ s}. \tag{3}$$

If the trial is the last question before the end of a session, then t_{ik}^{next} should be replaced by the end of the available continuous recording.

1.8 Why This Window Is Better Than a Fixed One

This window is preferable for Step 7 because:

- it respects the actual duration of the question-processing period for each trial;
- it incorporates a simple haemodynamic lag;
- it avoids contamination from the next trial by truncating at the next onset when necessary;
- it changes the interval length with the trial instead of forcing every trial into the same fixed window.

1.9 Baseline Definition

For each trial, define the baseline interval as:

$$B_{ik} = [t_{ik}^{\text{on}} - 2,\ t_{ik}^{\text{on}}]. \tag{4}$$

If the full two-second baseline is unavailable, the trial should be excluded from the primary analysis and logged.

1.10 Primary Trial-Level Outcome

Let $x_{ijk}(t)$ denote the preprocessed HbO signal for participant i, channel j, and trial k.

The baseline-corrected trial response for a channel is:

$$r_{ijk} = \overline{x_{ijk}(t \in W_{ik}^{(\delta)})} - \overline{x_{ijk}(t \in B_{ik})}. \tag{5}$$

The critical point is that this is the **mean** signal over the variable window, not the raw area. This avoids making longer trials mechanically larger simply because they last longer.

1.11 Front ROI Trial Summary

For each trial, the front ROI response is:

$$y_{ik}^{(F)} = \frac{1}{J_{ik}^{(F)}} \sum_{j \in F_{ik}} r_{ijk}, \tag{6}$$

where F_{ik} is the set of valid front-ROI channels available for participant i on trial k, and $J_{ik}^{(F)}$ is the number of such channels.

1.12 Back ROI Trial Summary

For secondary analyses, define:

$$y_{ik}^{(B)} = \frac{1}{J_{ik}^{(B)}} \sum_{j \in B_{ik}} r_{ijk}, \tag{7}$$

where B_{ik} is the set of valid back-ROI channels available for participant i on trial k.

1.13 Primary Covariates

Step 6 showed that the structural language variables were not strongly imbalanced by condition, but ENEM item correctness differed more substantially across abstract and concrete questions. Therefore, Step 7 should treat both question size and question difficulty explicitly.

The primary covariates should be:

- standardized log total word count;
- standardized ENEM item correctness.

Define:

$$w_k = z(\log(\texttt{total_word_count}_k)) \tag{8}$$

and

$$c_k = z(\texttt{correctness}_k). \tag{9}$$

1.14 Primary Model

Let:

$$A_{ik} = \begin{cases} 1 & \text{if the trial is Abstract} \\ 0 & \text{if the trial is Concrete} \end{cases} \tag{10}$$

The primary Step 7 model should be:

$$y_{ik}^{(F)} = \beta_0 + \beta_1 A_{ik} + \beta_2 w_k + \beta_3 c_k + u_i + v_k + \varepsilon_{ik}, \tag{11}$$

where:

- $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the participant random intercept;
- $v_k \sim \mathcal{N}(0, \sigma_v^2)$ is the question random intercept;
- $\varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2)$ is the residual term.

1.15 Primary Inferential Target

The primary inferential target is:

$$\beta_1, \tag{12}$$

which represents the adjusted abstract-versus-concrete effect in the left frontal ROI under the variable-window definition.

1.16 Primary Hypothesis

The primary hypotheses are:

Null hypothesis

$$H_0 : \beta_1 = 0 \tag{13}$$

Alternative hypothesis

$$H_1 : \beta_1 \neq 0 \tag{14}$$

The significance threshold should remain:

```
alpha = 0.05
```

1.17 Trial Inclusion Rules

A trial should be included in the primary Step 7 analysis if:

- it belongs to a participant-session already included in Step 6;
- it has a valid harmonized question onset;

- it has a valid harmonized answer time;
- the behavioral duration L_{ik} is positive;
- the two-second baseline interval is fully available;
- the final lagged variable window is fully available;
- the final lagged variable window length is at least 2.0 s after any truncation;
- at least one valid front ROI channel is present.

1.18 Participant Inclusion Rules

A participant-session should be retained in the primary Step 7 analysis if:

- it was included in Step 6;
- it contributes at least two valid abstract trials;
- it contributes at least two valid concrete trials;
- at least one valid front ROI channel remains for the included trials.

1.19 Pre-Model Duration Diagnostics

Before fitting the mixed models, Step 7 should include a descriptive analysis of how trial duration relates to condition and question properties.

At minimum, this pre-model diagnostic should report:

- response-time distribution by condition;
- correlation of response time with total word count;
- correlation of response time with sentence count;
- correlation of response time with sentence length;
- correlation of response time with ENEM item correctness.

This is useful because it quantifies the very mechanism that motivates the variable-window analysis.

1.20 Alternative Structure Models

To assess whether the result depends on the choice of structural covariate, Step 7 should also fit:

1.20.1 Alternative Model A: Sentence-count model

$$y_{ik}^{(F)} = \beta_0 + \beta_1 A_{ik} + \beta_2 s_k + \beta_3 c_k + u_i + v_k + \varepsilon_{ik}, \tag{15}$$

where

$$s_k = z(\texttt{sentence_count}_k). \tag{16}$$

1.20.2 Alternative Model B: Sentence-length model

$$y_{ik}^{(F)} = \beta_0 + \beta_1 A_{ik} + \beta_2 \ell_k + \beta_3 c_k + u_i + v_k + \varepsilon_{ik}, \tag{17}$$

where

$$\ell_k = z(\texttt{sentence_length}_k). \tag{18}$$

1.21 Secondary Back ROI Model

The same primary variable-window model should be fit for the back ROI as a secondary analysis:

$$y_{ik}^{(B)} = \beta_0 + \beta_1 A_{ik} + \beta_2 w_k + \beta_3 c_k + u_i + v_k + \varepsilon_{ik}. \tag{19}$$

1.22 Secondary Dissociation Model

To test whether the condition effect is more frontal than posterior under the variable-window definition, Step 7 should define a trial-level dissociation outcome:

$$d_{ik} = y_{ik}^{(F)} - y_{ik}^{(B)}. \tag{20}$$

The dissociation model should then be:

$$d_{ik} = \beta_0 + \beta_1 A_{ik} + \beta_2 w_k + \beta_3 c_k + u_i + v_k + \varepsilon_{ik}. \tag{21}$$

This is a secondary analysis.

1.23 Sensitivity Analyses

To keep Step 7 robust but still simple, the following sensitivity analyses should be included.

1.23.1 Sensitivity 1: Lag choice

Repeat the primary front ROI model using:

- $\delta = 3$ s;
- $\delta = 4$ s;
- $\delta = 5$ s.

1.23.2 Sensitivity 2: Cap long windows

Repeat the primary model using a capped lagged window:

$$W_{ik}^{(\delta,\mathrm{cap})} = \left[t_{ik}^{\mathrm{on}} + \delta,\ \min\!\left(t_{ik}^{\mathrm{ans}} + \delta,\ t_{ik}^{\mathrm{next}} - 0.5,\ t_{ik}^{\mathrm{on}} + 16\right)\right]. \tag{22}$$

This checks whether very long trials disproportionately influence the result.

1.23.3 Sensitivity 3: Response-time adjustment

Because response time is closely tied to the window length, it should not be included in the primary model. However, one sensitivity model may include standardized log response time:

$$r_{ik} = z(\log(L_{ik})) \tag{23}$$

and

$$y_{ik}^{(F)} = \beta_0 + \beta_1 A_{ik} + \beta_2 w_k + \beta_3 c_k + \beta_4 r_{ik} + u_i + v_k + \varepsilon_{ik}. \tag{24}$$

This model should be interpreted cautiously because response time may be partly downstream of condition.

1.23.4 Sensitivity 4: Fixed-window benchmark

Repeat the Step 6 primary front ROI model alongside the variable-window model so that the two approaches can be compared directly in the same report.

1.24 Step-by-Step Workflow

1.24.1 Step 7.1: Freeze the Step 6 cohort

Reuse the same included participant-session set as Step 6 unless a documented file or QC problem requires explicit re-review.

1.24.2 Step 7.2: Return to the continuous preprocessed data

Do not use the fixed-length Step 5/6 epochs as the primary input. Load the continuous preprocessed data from Step 3 together with the harmonized trigger table.

1.24.3 Step 7.3: Build the trial-level timing table

For each valid trial, store:

- participant ID;
- question ID;
- question condition;
- question onset time;
- answer time;
- next question onset time;
- raw answer latency;
- final lagged variable-window start;

- final lagged variable-window end;
- final variable-window duration.
- any truncation flag.

1.24.4 Step 7.4: Compute baseline-corrected ROI trial summaries

For each included trial:

- compute the front ROI mean over the variable window;
- compute the back ROI mean over the variable window;
- record the number of valid channels contributing to each ROI.

1.24.5 Step 7.5: Merge question covariates

Merge:

- total word count;
- sentence count;
- sentence length;
- ENEM item correctness.

1.24.6 Step 7.6: Run duration diagnostics

Produce descriptive plots and tables showing how response time varies by condition and by question properties.

1.24.7 Step 7.7: Fit the primary front ROI variable-window model

Fit the primary mixed-effects model:

- condition;
- log word count;
- ENEM correctness;
- random intercept for participant;
- random intercept for question.

1.24.8 Step 7.8: Fit the alternative structure models

Fit the sentence-count and sentence-length alternatives.

1.24.9 Step 7.9: Fit the secondary back and dissociation models

Fit the variable-window back ROI model and the variable-window dissociation model.

1.24.10 Step 7.10: Run the sensitivity analyses

Run:

- lag sensitivity;
- capped-window sensitivity;
- response-time sensitivity;
- fixed-window benchmark comparison.

1.25 Primary Reporting Quantities

The primary Step 7 results table must report:

- number of included participants;
- number of included questions;
- number of included trials;
- mean and distribution of variable-window durations;
- adjusted condition coefficient β_1;
- standard error;
- test statistic;
- p-value;
- 95% confidence interval;
- word-count coefficient;
- ENEM-correctness coefficient;
- participant random-intercept variance;
- question random-intercept variance.

1.26 Secondary Reporting Quantities

The secondary Step 7 tables should report:

- back ROI variable-window model results;
- dissociation model results;
- sentence-count alternative model results;
- sentence-length alternative model results;
- lag sensitivity results;
- capped-window sensitivity results;
- response-time sensitivity results;
- fixed-window benchmark comparison.

1.27 Primary Conclusion Rule

The final Step 7 conclusion should be based first on the condition coefficient β_1 from the primary variable-window front ROI model.

The interpretation rule should be:

- if the adjusted condition coefficient is significant and positive, conclude that abstract questions are associated with a larger frontal HbO response than concrete questions even when the response is summarized over a lagged question-specific duration-aware interval;
- if the adjusted condition coefficient is significant and negative, conclude that concrete questions are associated with a larger frontal HbO response than abstract questions under the variable-window definition;
- if the adjusted condition coefficient is not significant, conclude that the variable-window adjusted model does not show statistically significant evidence for a frontal abstract-versus-concrete effect.

1.28 Conclusion Templates

Template if the primary effect is significant and abstract > concrete. In the duration-aware mixed-effects analysis, abstract questions were associated with a significantly larger left frontal ROI HbO response than concrete questions after adjusting for log word count and ENEM item correctness ($\beta = \ldots$, $SE = \ldots$, $p = \ldots$, 95% CI $[\ldots, \ldots]$). This indicates that the frontal condition effect remains when the analysis window is allowed to vary with the actual duration of each question.

Template if the primary effect is significant and concrete > abstract. In the duration-aware mixed-effects analysis, concrete questions were associated with a significantly larger left frontal ROI HbO response than abstract questions after adjusting for log word count and ENEM item correctness ($\beta = \ldots$, $SE = \ldots$, $p = \ldots$, 95% CI $[\ldots, \ldots]$).

Template if the primary effect is not significant. In the duration-aware mixed-effects analysis, the abstract-versus-concrete effect in the left frontal ROI was not statistically significant after adjusting for log word count and ENEM item correctness ($\beta = \ldots$, $SE = \ldots$, $p = \ldots$, 95% CI $[\ldots, \ldots]$). This suggests that allowing the trial window to vary with question duration does not by itself yield statistically significant evidence for a frontal condition effect in the current dataset.

1.29 Expected Outputs

Tables

```
01_step7_trial_timing_table.csv
02_step7_duration_diagnostics.csv
03_step7_trial_variable_window_summary.csv
04_step7_primary_front_variable_window_model.csv
05_step7_front_sentencecount_model.csv
06_step7_front_sentencelength_model.csv
07_step7_back_variable_window_model.csv
08_step7_dissociation_variable_window_model.csv
09_step7_lag_sensitivity.csv
10_step7_capped_window_sensitivity.csv
11_step7_rt_sensitivity.csv
12_step7_fixed_vs_variable_benchmark.csv
13_step7_exclusion_log.csv
14_step7_final_conclusion.csv
```

Figures

```
figures/step7/response_time_by_condition.png
figures/step7/response_time_vs_wordcount.png
figures/step7/response_time_vs_correctness.png
figures/step7/variable_window_duration_histogram.png
figures/step7/front_roi_partial_effect_variable_window.png
figures/step7/back_roi_partial_effect_variable_window.png
figures/step7/dissociation_partial_effect_variable_window.png
figures/step7/fixed_vs_variable_comparison.png
figures/step7/model_residuals_primary.png
```

Reports

```
reports/step7/step7_variable_window_report.tex
reports/step7/tables/primary_model_table.tex
reports/step7/tables/secondary_model_table.tex
reports/step7/text/final_conclusion.tex
```

1.30 Minimal Figures for the Main Report

The main Step 7 report should include at least:

1. response-time distribution by condition;
2. response time versus total word count;
3. response time versus ENEM item correctness;
4. histogram of final variable-window durations;
5. partial-effect plot for the primary front ROI variable-window model;
6. direct benchmark comparison of the Step 6 fixed-window coefficient and the Step 7 variable-window coefficient.

1.31 Quality-Control Checks

Before Step 7 is considered complete, verify that:

1. the Step 6 cohort is unchanged unless explicitly re-reviewed;
2. the continuous preprocessed files are used rather than truncated onset epochs;
3. every included trial has valid onset and answer times;
4. every included trial has a fully available baseline interval;
5. the variable window does not overlap the next trial onset;
6. the final variable window length is positive and at least 2 s;
7. only long-separation channels are used in the inferential ROI summaries;
8. the short-separation pairs remain excluded from inference;
9. the front ROI channel list is identical to Step 6;
10. the primary model includes both question size and question difficulty;
11. the primary model converges;
12. all sensitivity analyses are clearly labeled as non-primary.

1.32 Fallback Rule

If the primary mixed-effects model fails to converge reliably, the fallback analysis should be:

> a linear model with participant fixed effects and question fixed effects, keeping the same variable-window front ROI outcome and the same primary covariates.

1.33 Minimal Completion Criteria

Step 7 is complete when:

- the variable trial windows have been constructed;
- the duration diagnostics have been completed;
- the primary front ROI variable-window model has been fit;
- the alternative structure models have been fit;
- the secondary back and dissociation models have been fit;
- the lag and capped-window sensitivities have been completed;
- the fixed-window versus variable-window benchmark comparison has been completed;
- the primary adjusted p-value has been reported;
- a short final conclusion has been generated;
- all exclusions and warnings have been documented.

1.34 Short Practical Summary

In simple terms, Step 7 should:

> reuse the same cohort and the same frontal ROI from Step 6, move from a fixed 7–11 s summary to a lagged question-specific variable window that runs with the actual duration of each trial, adjust for both question length and ENEM difficulty, and test whether the abstract-versus-concrete effect survives under this more duration-aware definition.